

FitBuddy – AI Fitness Plan Generator

5. Project Development Phase

Core Development

The project is developed using FastAPI, Gemini models, Jinja2 templates, SQLAlchemy, and SQLite.

Workout Generation

`generate_workout_gemini()` uses user goal and intensity to create a structured 7-day plan with warm-up, main workout, and cooldown/recovery guidance.

Nutrition Tip

`generate_nutrition_tip_with_flash()` provides a concise nutrition or recovery suggestion related to the selected goal.

Feedback Update

`update_workout_plan()` sends the original plan and user feedback to Gemini Pro and produces a revised plan.

Database

Functions such as `save_user()`, `save_plan()`, `update_plan()`, `get_original_plan()`, and `get_user()` support storage and retrieval.

Routing

`routes.py` connects form input, AI generation functions, database operations, and HTML templates.

Frontend

`index.html`, `result.html`, and `all_users.html` provide the main user and administrator interfaces.

Project: FitBuddy – AI Fitness Plan Generator using Gemini Models.